
Shunting Concentrates Path Gains and Improves Local Credit Assignment in Dendritic Networks

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Abstract

1 Exact gradients for conductance-based dendritic trees factorize into a synapse-local
2 eligibility term and a single non-local compartment error; this factorization is the
3 foundation for local credit assignment in these networks. How well a somatic broad-
4 cast can substitute for the true compartment error depends on how concentrated
5 the path gains are across the tree. An oracle experiment—replacing the broadcast
6 with the exact transported compartment errors—raises additive local learning from
7 33% to 98.5% on noise resilience, nearly matching shunting (98.7%), strongly sup-
8 porting path-gain fidelity as the dominant bottleneck rather than other architectural
9 differences. Under simple per-soma broadcast, shunting improves both gradient
10 fidelity and end-task learning relative to additive integration: cosine alignment
11 rises from 0.006 to 0.202, and noise-resilience accuracy rises from 0.33 to 0.90.
12 The advantage is regime-dependent, growing with inhibitory conductance and
13 largest under noisy inputs. A structured pathway-vector extension (PV-LocalCA)
14 further stabilizes learned-routing performance on a hard cue-integration benchmark,
15 reaching 99.1% test accuracy. Together, these results identify path-gain concen-
16 tration via shunting inhibition as the key biophysical mechanism that determines
17 when low-bandwidth local credit assignment works in conductance-based dendritic
18 networks.

19 1 Introduction

20 Credit assignment in deep networks usually depends on backpropagation: global error transport
21 through symmetric pathways with no obvious biological substrate. Dendritic compartments suggest an
22 alternative. Each synapse has access to rich local state such as driving force, membrane conductance,
23 and branch voltage, while supervisory information could arrive as a low-bandwidth broadcast from
24 the soma [15]. The question here is when such local information is sufficient for useful learning.

25 Starting from conductance-based dendritic voltage equations [1], we derive exact gradients for
26 dendritic trees (Theorem 1) and show that each synaptic gradient factorizes into purely local eligibility
27 terms and a single non-local compartment error. Approximating that compartment error with a
28 broadcast signal yields a theorem-facing local rule (3F), together with practical heuristic wrappers
29 (4F/5F), and a structured pathway-vector extension for tasks where different branches must be credited
30 differently.

31 The central finding is mechanistic: shunting improves broadcast fidelity by concentrating path
32 gains. On MNIST, path-gain CV stays at 0.21–0.27 under shunting versus 0.53–1.08 under additive
33 integration, so one shared broadcast is a much better proxy for compartment-specific errors under
34 shunting. On the noise-resilience sweep, per-soma broadcast achieves cosine alignment of about
35 0.20–0.32 under shunting, but stays near zero or negative under additive integration. Our question
36 is therefore not whether local learning can match backpropagation on standard benchmarks, but

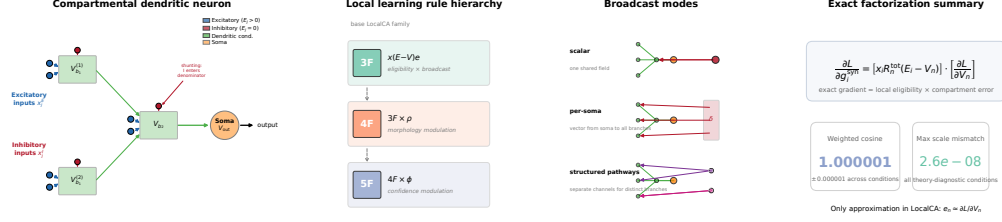


Figure 1: Model and credit assignment. (A) A compartmental dendritic neuron with excitatory (blue triangles) and inhibitory (red circles; shunting) synapses. Voltages propagate to the soma through learned dendritic conductances (green arrows). A somatic broadcast error e_n (dashed red) reaches all compartments; each synapse combines e_n with local quantities (presynaptic drive x_j , input resistance R_n^{tot} , and driving force $E_j - V_n$). (B) Compact LocalCA hierarchy: 3F is the theorem-facing local rule; 4F adds the heuristic morphology-aware factor ρ ; 5F further adds the heuristic confidence-like factor ϕ . (C) Broadcast-mode schematic. Scalar broadcast sends one shared field to all branches; per-soma broadcast sends a vector-valued somatic error to each branch; structured pathway feedback uses separate channels to preserve pathway identity when different branches should receive different teaching signals. (D) Headline exact-factorization summary. The panel states the factorization directly and reports the verification numbers compactly; Fig. 2D then shows the full diagnostic scatter across all theory-diagnostic conditions.

what biophysical mechanism determines when low-bandwidth local credit assignment works in conductance-based dendritic networks. An oracle experiment that supplies the exact transported compartment errors strongly supports path-gain fidelity as the dominant bottleneck: additive local learning rises to 98.5% on noise resilience under the oracle, nearly matching shunting (98.7%).

We quantify these effects with a gradient-fidelity diagnostic based on directional alignment and scale mismatch between local and backprop gradients (Fig. 2). The absolute cosines remain modest, but they are consistently more positive under shunting and predict which local-learning regimes ultimately succeed. The paper’s contributions are: exact factorization for conductance-based dendritic trees, a mechanistic account of shunting as path-gain concentration, direct fidelity and oracle experiments that isolate the broadcast bottleneck, and a structured rank- K extension for settings where rank-1 broadcast fails.

2 Compartmental Voltage Model and Gradient Derivation

We use a steady-state conductance model derived from discretized passive cable dynamics [1, 2]. In normalized units, leak and inhibitory reversal are set to 0, the excitatory reversal is set to 1, and leak conductance is fixed at 1. Each compartment voltage is therefore a conductance-weighted average. Two properties of this form matter for what follows: local sensitivities depend on both driving force $(E - V)$ and input resistance R^{tot} , and shunting inhibition corresponds to adding conductance when $E_{\text{inh}} \approx 0$.

2.1 Voltage Equation and Local Sensitivities

Consider compartment n with synaptic inputs j (activity x_j , reversal E_j , conductance $g_j^{\text{syn}} \geq 0$) and dendritic inputs from children (voltage V_j , conductance $g_j^{\text{den}} \geq 0$). The steady-state voltage is:

$$V_n = \frac{\sum_j E_j x_j g_j^{\text{syn}} + \sum_j V_j g_j^{\text{den}}}{\underbrace{\sum_j x_j g_j^{\text{syn}} + \sum_j g_j^{\text{den}} + 1}_{g_n^{\text{tot}}}}, \quad R_n^{\text{tot}} = 1/g_n^{\text{tot}}. \quad (1)$$

V_n is a convex combination of reversal potentials, child voltages, and leak. Writing $\mathcal{S}_n = \{0\} \cup \{E_j\}_j \cup \{V_j\}_j$, we have $\min \mathcal{S}_n \leq V_n \leq \max \mathcal{S}_n$ and $0 < R_n^{\text{tot}} \leq 1$. The local sensitivities follow directly:

Proposition 1 (Local Sensitivities).

$$\frac{\partial V_n}{\partial g_i^{\text{syn}}} = x_i R_n^{\text{tot}} (E_i - V_n), \quad \frac{\partial V_n}{\partial V_i} = g_i^{\text{den}} R_n^{\text{tot}}, \quad \frac{\partial V_n}{\partial g_i^{\text{den}}} = R_n^{\text{tot}} (V_i - V_n). \quad (2)$$

2.2 Shunting Inhibition as Divisive Gain Control

An inhibitory synapse with $E_{\text{inh}} \approx 0$ contributes current $(0 - V_n)x_j g_j^{\text{syn}}$ and increases g_n^{tot} . Its sensitivity is $\partial V_n / \partial g_j^{\text{syn}} = -x_j R_n^{\text{tot}} V_n$, which corresponds to multiplicative attenuation (divisive normalization). Shunting is divisive at the voltage level, but its effect on firing rates can be subtractive in some regimes [7]. Inhibitory plasticity can balance excitation dynamically [8]; our learned inhibitory conductances serve an analogous role.

2.3 Exact Gradients for Dendritic Trees

Let V_0 be the somatic output, $\hat{y} = W_{\text{dec}} V_0$ the decoder, and $\delta_0 = W_{\text{dec}}^\top (\partial L / \partial \hat{y})$ the somatic error.

Theorem 1 (Backpropagation on a Dendritic Tree). *For a rooted dendritic tree with soma at node 0 and unique parent $p(n)$ for each non-somatic compartment, the loss gradient satisfies*

$$\frac{\partial L}{\partial V_n} = \frac{\partial L}{\partial V_{p(n)}} R_{p(n)}^{\text{tot}} g_{n \rightarrow p(n)}^{\text{den}}, \quad (3)$$

with boundary condition $\partial L / \partial V_0 = \delta_0$. Because the graph is a tree, each compartment has a unique path to the soma. Defining the path gain

$$\alpha_n = \prod_{(i \rightarrow k) \in \text{path}(n \rightsquigarrow 0)} R_k^{\text{tot}} g_{i \rightarrow k}^{\text{den}}, \quad \frac{\partial L}{\partial V_n} = \alpha_n \delta_0, \quad (4)$$

with $\alpha_0 = 1$.

Proof. Apply the chain rule on the tree-structured computation graph using Prop. 1. $\square$

Corollary 1 (Local–Global Factorization). *The exact synaptic gradient at compartment n factorizes as:*

$$\frac{\partial L}{\partial g_i^{\text{syn}}} = \underbrace{x_i R_n^{\text{tot}} (E_i - V_n)}_{\text{synapse-local eligibility}} \cdot \underbrace{\frac{\partial L}{\partial V_n}}_{\text{compartment error}}, \quad (5)$$

where the eligibility term depends only on quantities available at synapse i (presynaptic activity x_i , input resistance R_n^{tot} , driving force $E_i - V_n$), and the compartment error $\partial L / \partial V_n$ is the sole non-local quantity.

This factorization isolates the only non-local quantity: $\partial L / \partial V_n = \alpha_n \delta_0$. Any practical local rule therefore depends entirely on how well a broadcast signal approximates that exact compartment error, which we evaluate in Sec. 4.2. In the models studied here, shunting improves this approximation by reducing and stabilizing R_n^{tot} , which narrows the spread of local sensitivities across the tree.

3 Local Learning Rules

3.1 Broadcast Error Approximation

We approximate the exact compartment error $\partial L / \partial V_n$ (Corollary 1) with a broadcast signal e_n derived from the somatic error $\delta_0 = W_{\text{dec}}^\top (\partial L / \partial \hat{y})$. We consider five broadcast modes of increasing structure:

- **Scalar:** $e_n = \bar{\delta} \cdot 1$, where $\bar{\delta} = \text{mean}(\delta_0)$ is a single scalar broadcast to all compartments.
- **Per-soma:** $e_n = \delta_0$ when the layer dimension matches the output; layers with mismatched dimensions fall back to scalar.
- **Low-rank:** $e_n = \Gamma_K(\delta_0)$, where Γ_K uses a small number K of random broadcast channels to project the somatic error into a layer-specific vector field.

- **Local mismatch:** $e_n = (1-\alpha) \bar{\delta} \cdot \tilde{m}_n + \alpha \bar{\delta}$, where $\tilde{m}_n$ is the RMS-normalized, batch-centered parent-child voltage difference $P_n - V_n$ and $\alpha = 0.2$.
- **Pathway-vector:** $e_n = \alpha \bar{\delta} \cdot \mathbf{1} + (1-\alpha) \Gamma_n(\delta_0)$, where Γ_n projects the somatic error into router-inferred pathway roles, gates those roles by local pathway activity, and transports the resulting vector hierarchically through the block-structured dendritic tree.

We use per-soma broadcast by default. In our experiments, the local-mismatch variant performs much worse (Appendix A), indicating that broadcast quality is critical. Under shunting, this simple broadcast is often sufficient on standard classification tasks, but the cue-routing benchmark in Sec. 4.7 requires the structured pathway-vector form. Appendix Table S8 summarizes the broader set of implemented model families, training strategies, and broadcast modes used throughout the study.

3.2 Three-Factor Rule (3F)

Definition 1 (3F Update). *For synaptic and dendritic conductances:*

$$\Delta g_j^{\text{syn}} = \eta \langle x_j R_n^{\text{tot}} (E_j - V_n) e_n \rangle_B, \quad \Delta g_j^{\text{den}} = \eta \langle R_n^{\text{tot}} (V_j - V_n) e_n \rangle_B, \quad (6)$$

where $\langle \cdot \rangle_B$ denotes the batch average.

The three factors are presynaptic activity x_j (or a voltage difference), postsynaptic modulation through driving force and input resistance, and the broadcast error e_n . The same rule applies to excitatory and inhibitory synapses. The sign difference comes from the driving force $(E_j - V_n)$.

Additive control. Rule (6) is the local gradient of the *shunting* voltage $V_n = \sum E_j g_j x_j / g_n^{\text{tot}}$. For the additive control ($V_n = \sum g_j x_j$, no divisive normalization), the correct local gradient is simpler: $\Delta g_j^{\text{syn}} = \eta \langle x_j e_n \rangle_B$, with no driving-force or R_n^{tot} terms ($R_n^{\text{tot}} \equiv 1$ by definition since there is no denominator). Throughout, each architecture uses the learning rule derived from its own forward-pass dynamics. This keeps the comparison architecture-matched rather than applying a shunting-derived rule to an additive model.

3.3 Practical Heuristic Wrappers: 4F and 5F

4F (heuristic attenuation proxy). In the exact gradient (4), the path gain α_n attenuates the somatic error differently at each compartment. To compensate without explicitly computing that gain, 4F uses an online correlation proxy

$$\rho_n = \text{Cov}(\bar{V}_n, \bar{V}_0) / (\sqrt{\text{Var}(\bar{V}_n) \text{Var}(\bar{V}_0)} + \varepsilon).$$

High ρ_n up-weights branches whose voltages covary with the soma; low ρ_n down-weights branches where the broadcast is likely to be a poor proxy.

5F (confidence-weighted 4F). 5F adds a second heuristic factor

$$\phi_n = \text{Var}(V_n) / (\sigma_{\text{res}}^2 + \varepsilon),$$

where σ_{res}^2 is the residual variance of V_n after linear regression on the parent voltage. Operationally, ϕ_n is a branch-wise confidence gate: it becomes large when local voltages are strong and stable relative to unexplained residual fluctuations, and small when local fluctuations dominate. We clamp it to $[0.25, 4.0]$ for stability. This factor is empirical, not theorem-derived. The paper’s core claim rests on path-gain concentration and the oracle experiment; the gain from ϕ_n should be read as practical denoising rather than theoretical necessity.

Proposition 2 (5F Update).

$$\Delta g_j^{\text{syn}} = \eta \rho_n \phi_n \langle x_j R_n^{\text{tot}} (E_j - V_n) e_n \rangle_B. \quad (7)$$

A feedback-alignment-style random-broadcast argument is included in Appendix C; it is supportive but not central to the main mechanism claim.

3.4 Structured pathway feedback and branch-role plasticity

For tasks with latent pathway structure, scalar broadcast can erase the distinction between different dendritic branches that should be credited differently. Recent experimental evidence supports the existence of vectorized instructive signals in cortical dendrites [47], motivating the structured feedback variant we introduce here. We therefore augment the local rule with a structured cue-routing variant, *PV-LocalCA*.

Hierarchical pathway-vector broadcast. The encoder/router defines a low-dimensional role basis over latent pathways. Instead of collapsing δ_0 to a scalar or per-soma vector, we project it into this role basis and gate each role by local pathway activity:

$$e_n^{\text{PV}} = \sum_{k=1}^K q_{n,k} c_k,$$

where $c_k = \langle r_k, \delta_0 \rangle$ are role coefficients and $q_{n,k}$ are local branch-role weights. Scalar broadcast is the special case $K=1$, while PV-LocalCA is a structured rank- K extension rather than a separate learning principle.

Router-derived branch roles. Rather than assigning hand-coded apical/basal labels, each branch j receives a normalized role profile r_j inferred from router assignments and incoming block weights. Plasticity is then gated by branch selectivity and by a synapse-specific role-alignment factor $\langle r_i, r_j \rangle$, biasing learning toward pathway-consistent afferents without imposing a fixed branch taxonomy.

4 Experiments

4.1 Setup

We evaluate two settings: a *capacity-calibrated* regime, where matched architectures achieve strong backprop performance, and *controlled sweeps* that isolate inhibition strength, broadcast mode, and architecture. Our main datasets are MNIST, Fashion-MNIST [40], and four synthetic tasks: *context gating*, *noise resilience*, *info shunting*, and *cue integration*; formal definitions are in Appendix I. Architectures include point MLP baselines and dendritic cores with either additive integration or shunting. Main comparisons are architecture-matched additive vs. shunting cores and local vs. backprop training; DFA appears only as a supplementary baseline. Most headline MNIST, Fashion-MNIST, and context-gating results use 5F with per-soma broadcast. The cue-routing benchmark instead uses PV-LocalCA, and the inhibition sweeps additionally use a transported-error oracle derived from Theorem 1 as an upper bound rather than as the proposed biological rule. Headline mechanistic and routed results use 3–5 seeds; 1–2 seed screens are treated as exploratory and are either omitted from the main claims or explicitly labeled as such in the appendix. The experiments follow one chain: first verify how closely different broadcasts track the exact gradients implied by the theorem, then test whether those fidelity differences translate into learning, and finally study a structured-feedback extension for settings where rank-1 broadcast fails.

4.2 From Exact Factorization to Credit-Signal Fidelity

To test whether these performance differences correspond to better credit signals, we compare local and backprop gradients on the same batch at the same parameter values. For each parameter tensor p , we compute directional alignment (cosine similarity) and scale mismatch, defined as $|\log_{10}(\|g_p^{\text{local}}\|/\|g_p^{\text{bp}}\|)|$, aggregated by parameter count.

Across matched-weight comparisons, shunting is consistently more aligned with backprop and substantially closer in norm than the additive control: on MNIST, weighted cosine rises from 0.006 to 0.202 and scale mismatch falls from 1.053 to 0.117; on context gating, cosine rises from -0.007 to 0.108 and scale mismatch falls from 2.154 to 0.036. Figure 2C shows that this advantage appears throughout training, especially in proximal layers and dendritic conductances. Figure 2D confirms that the theory is operational in code: reconstructing gradients from Corollary 1 using exact compartment errors matches autograd to numerical precision (weighted cosine 1.000001 ± 0.000001 , weighted scale mismatch $< 4 \times 10^{-8}$). The only approximation made by LocalCA is therefore the broadcast substitute for $\partial L / \partial V_n$.

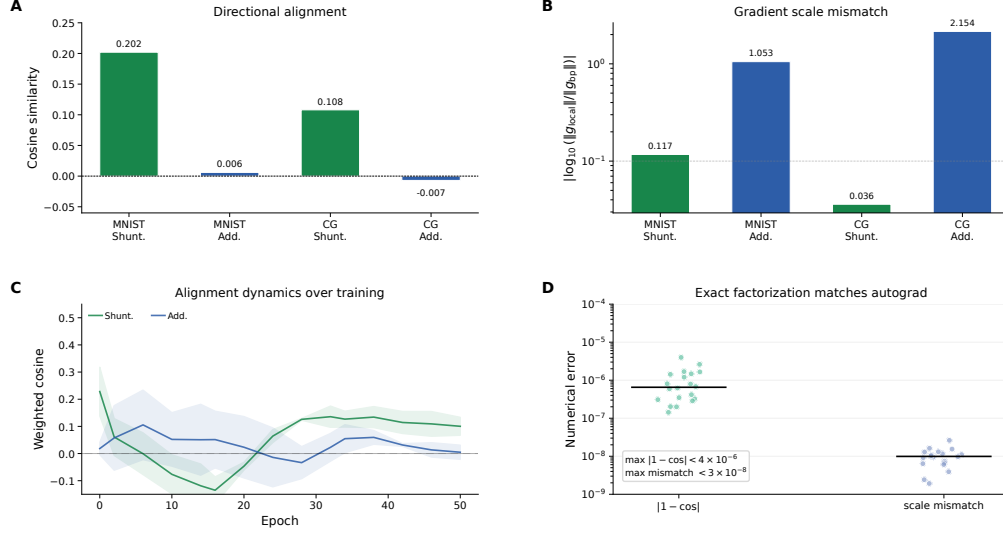


Figure 2: Gradient-fidelity diagnostic. (A) Weighted cosine similarity between local and backprop gradients: shunting (green) is consistently more aligned than additive (blue) on both MNIST and context gating. (B) Gradient scale mismatch on the same matched-weight comparisons. Shunting is not only better aligned, but also substantially closer in norm. (C) Per-layer cosine similarity over training epochs. Shunting proximal layers approach high positive alignment; additive layers remain near zero. (D) Exact-factorization sanity check across all theory-diagnostic conditions. Reconstructed gradients match autograd to numerical precision, showing that LocalCA’s only approximation is in the broadcast substitute for the compartment error.

178 4.3 Mechanistic Chain: Path Gains, Broadcast Fidelity, and Oracle Transport

179 Figure 3 provides the paper’s central empirical chain: shunting concentrates path gains, which
 180 improves direct compartment-error fidelity, which in turn makes low-bandwidth local learning
 181 effective; the transported-error oracle then isolates how much of the remaining gap is due to broadcast
 182 quality rather than other architectural differences.

183 Panel A measures the coefficient of variation of the exact path gains α_n across compartments. On
 184 MNIST, shunting keeps this distribution tightly concentrated across inhibition levels (CV about
 185 0.21–0.27), whereas the additive control remains much broader (CV about 0.53–1.08). Panel B
 186 then measures the sole non-local term from Corollary 1, $\partial L/\partial V_n$, directly. On the noise-resilience
 187 sweep, per-soma broadcast tracks the exact compartment error much better under shunting than
 188 under additive integration, rising from about 0.20 at $N_I=0$ to about 0.32 near the best-performing
 189 inhibition range, while additive stays near zero or negative. The transported-error oracle obtained by
 190 recursively applying exact path gains improves fidelity further in both architectures, showing that the
 191 theorem defines an operational upper bound on low-bandwidth broadcast.

192 Panel C trains the same local rule using that oracle signal. On noise resilience, additive learning rises
 193 from 33–47% under per-soma broadcast to 75.6% at $N_I=0$ and 98.5% at $N_I=40$ under transported
 194 broadcast. Shunting improves from 71.7–90.3% under per-soma to 98.5–99.3% across the same
 195 sweep. On MNIST, transported broadcast reaches 94.6% for additive and 96.8% for shunting. These
 196 oracle results strongly support path-gain fidelity as the dominant bottleneck for LocalCA performance,
 197 while the smaller remaining additive–shunting gap under matched transported broadcast suggests an
 198 additional contribution from conductance-shaped local sensitivities.

199 Panel D shows that coarse broadcast can still be enough once local sensitivities are well scaled: 4-bit
 200 quantization preserves full performance (61.6% vs. 61.7%; $p > 0.9$), while sign-only (37.5%) and
 201 sparse top-30% (38.2%) degrade sharply. Normalization-only additive controls and DFA baselines
 202 (Appendix Figs. S9, S7) improve over weaker baselines without matching the shunting conductance
 203 model.

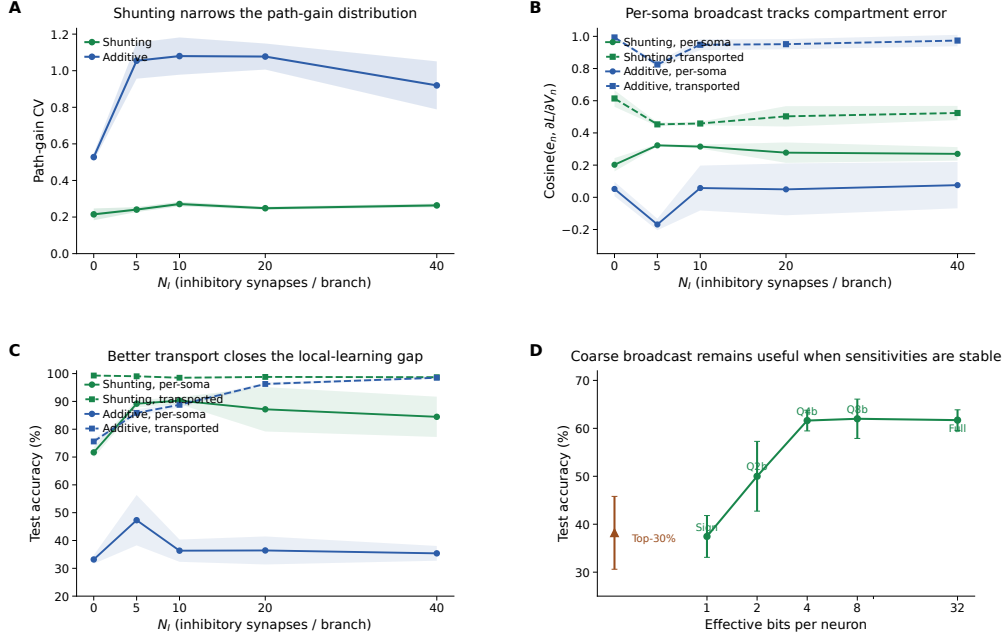


Figure 3: **Mechanistic evidence.** (A) Path-gain dispersion on MNIST, measured as the coefficient of variation of the exact tree attenuation α_n across compartments. Shunting keeps path gains much more concentrated than the additive control as inhibition increases. (B) Direct compartment-error fidelity on noise resilience. Solid lines compare the per-soma broadcast e_n to the exact compartment error $\partial L / \partial V_n$; dashed lines show the transported-error oracle obtained by recursively applying exact path gains. Shunting substantially improves the fidelity of the simple per-soma broadcast, while the transported oracle defines the remaining approximation gap. (C) Learning with the transported-error oracle on noise resilience. Improving the broadcast field is enough to recover near-ceiling local learning, especially in the additive architecture where per-soma broadcast fails most severely. (D) Broadcast bandwidth. Four-bit quantization preserves full performance; only sign-only and sparse modes degrade sharply.

204 4.4 Capacity-Calibrated Competence

205 Having established the mechanism, we next ask whether the same conditions improve end-task
 206 learning in architectures that are already competent under backpropagation. In the capacity-calibrated
 207 regime, shunting dendritic cores remain reasonably competitive under local learning: on MNIST,
 208 Fashion-MNIST, and context gating, the best local configuration (5F, per-soma broadcast, local
 209 decoder) stays within about 5–7 percentage points of the matched backprop ceiling (Appendix
 210 Table S4; Appendix Fig. S2). Within the local-rule family, 5F consistently outperforms 4F and
 211 3F (Appendix Table S1), and per-soma strongly outperforms scalar and local-mismatch broadcast
 212 (Appendix Table S3). We therefore treat 5F as the strongest practical wrapper in this paper, not as a
 213 uniquely derived optimum; its clamp bounds and residual estimator remain heuristic design choices,
 214 although the main tuning knobs are not especially fragile (Appendix Fig. S4). Appendix Table S2
 215 summarizes the empirical division of labor across architecture, rule, and broadcast choices.

216 4.5 Regime Dependence of the Shunting Advantage

217 The benefit of shunting varies across tasks. On MNIST with matched per-soma broadcast, shunting
 218 outperforms additive by about 2 percentage points; a similar pattern holds on Fashion-MNIST
 219 (81.1% vs. 79.4%). On tasks that require credit signals robust to noise, the gap is much larger:
 220 +50.3 pp on noise resilience (with $N_I=10$ inhibitory synapses per branch) and +24.8 pp on info
 221 shunting ($N_I=0$). Appendix Fig. S2 shows the full dose-response: the shunting advantage grows
 222 with inhibitory conductance strength on the inhibition-sensitive tasks. Additive cores can still perform
 223 reasonably on easier settings, reaching about 89% on MNIST after tuning, but they fail in regimes
 224 where inhibition-mediated normalization matters for gradient propagation.

225 4.6 Additional Stress Tests

226 Across three additional stress tests — increased dendritic depth, noisy broadcast signals, and Fashion-
227 MNIST — the same qualitative pattern persists: shunting degrades more gracefully than additive,
228 while the largest gains still appear on tasks that stress the credit signal most strongly. We keep the
229 full curves in Appendix Fig. S6 because they support, rather than define, the core mechanistic claim.

230 4.7 Extension: Structured Feedback When Per-Soma Broadcast Fails

231 The main paper focuses on low-bandwidth scalar or per-soma broadcast. We now consider a harder
232 cue-integration benchmark where that approximation is insufficient because different dendritic
233 pathways must be credited differently. The task asks the model to combine two noisy sensory cues
234 using a context signal that indicates which cue is reliable on each trial. In the fixed-pathway control,
235 context is copied into both cue pathways. In the learned-pathway setting, the router must instead
236 discover that separation and align one latent pathway with each cue stream.

237 Figure 4 summarizes five-seed runs for this benchmark. The benchmark itself is learnable: fixed-
238 pathway additive local learning remains near ceiling ($99.1 \pm 0.3\%$), learned-routing additive local
239 learning also succeeds ($98.1 \pm 2.4\%$), and standard learned-router backprop training stays near
240 $99.2 \pm 0.4\%$ for both additive and shunting cores. The specific failure appears when learned-routing
241 shunting LocalCA uses the default per-soma broadcast. At the original 90-epoch setting, accuracy is
242 only $82.6 \pm 14.2\%$, and modest router or optimization changes help only slightly: low temperature
243 gives $87.6 \pm 13.3\%$, freeze-20 gives $83.8 \pm 13.9\%$, and a tuned per-soma run reaches $88.2 \pm 13.9\%$.

244 Replacing per-soma broadcast with PV-LocalCA changes the regime. The untuned 90-epoch version
245 improves to $94.0 \pm 11.0\%$, and the tuned setting (140 epochs, learning rate 10^{-3}) reaches $99.1 \pm$
246 0.3% test and $99.4 \pm 0.2\%$ validation accuracy while maintaining strong specialization (mean max
247 assignment 0.995 ± 0.008). The failure is therefore not pathway discovery itself; it is the loss of
248 pathway identity under rank-1 feedback. Consistent with that interpretation, simply increasing the
249 number of random low-rank channels does not recover the same gain (Appendix Table S6). When
250 the feedback preserves that structure, learned-routing shunting LocalCA becomes reliable.

251 5 Related Work

252 **High-performance local learning.** Recent methods such as AugLocal, CCL, and DLL have
253 substantially raised the benchmark-performance bar for local learning on standard architectures
254 [36, 37, 39]. Our aim is therefore not to claim the strongest benchmark accuracy among local-
255 learning methods, but to identify the biophysical mechanism that determines when low-bandwidth
256 credit signals become useful in conductance-based dendritic networks.

257 **Local credit-assignment dynamics.** Random feedback, DFA, DTP, DFC, equilibrium-style meth-
258 ods, and predictive-coding approaches solve locality through different feedback or inference dynamics
259 [9, 10, 14, 16, 17, 27–31, 35]. EBD [38] is especially close in spirit because it studies when di-
260 rect error broadcast can be made useful by shaping the broadcast field itself. Our contribution is
261 complementary: rather than changing the broadcast objective or inference dynamics, we identify
262 a conductance-based inductive bias—path-gain concentration under shunting—that makes even a
263 simple low-bandwidth broadcast more faithful to the exact compartment error.

264 **Dendrites for robustness and context.** Dendritic trees support nonlinear computation [3, 4, 32]
265 and have inspired learning schemes based on segregated dendrites, dendritic prediction errors, burst-
266 dependent plasticity, and latent equilibrium [5, 11, 12, 24–26]. Outside supervised iid classification,
267 active-dendrite architectures can also improve robustness in dynamic multi-task settings where
268 point-MLP baselines suffer interference [48]. Our cue-routing extension asks a related question—
269 when dendritic structure helps preserve pathway-specific computation—but uses conductance-based
270 normalization and structured credit signals rather than sparse kWTA gating.

271 **Neuroscience grounding.** Shunting implements divisive normalization [6], though its rate effects
272 can be subtractive in some regimes [7]; inhibitory conductance also modulates gain and signal-
273 to-noise [33]. Recent evidence for vectorized instructive signals in cortical dendrites [47] further

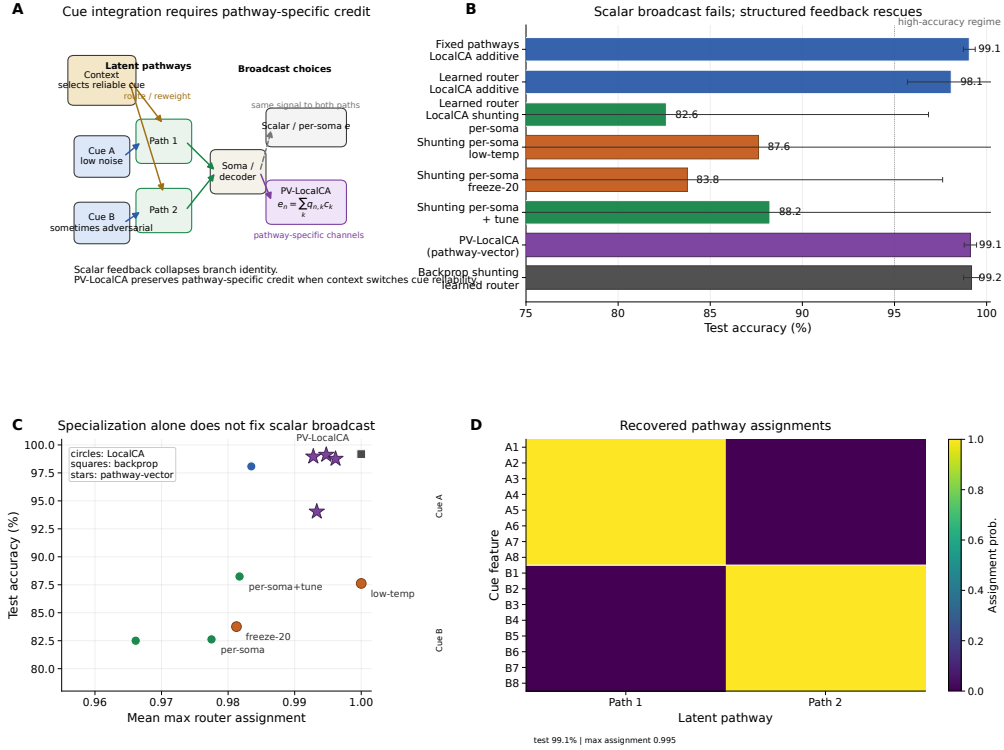


Figure 4: Structured feedback for pathway discovery on cue integration. (A) Task schematic. Context determines which noisy cue is reliable on each trial. Scalar or per-soma broadcast sends the same teaching signal to both pathways, whereas PV-LocalCA preserves pathway identity through a low-rank pathway-vector feedback field. (B) Seed-averaged diagnosis and focused tuning (5 seeds per condition). Learned-routing additive local learning remains high, but learned-routing shunting with per-soma broadcast stays unstable; low-temperature, freeze-20, and per-soma tuning help only modestly, whereas tuned pathway-vector feedback reaches $99.1 \pm 0.3\%$. (C) Test accuracy vs. router specialization across learned-router conditions. High specialization alone does not guarantee success: per-soma, low-temperature, and freeze-20 runs remain inaccurate despite near-discrete assignments. (D) Recovered pathway assignments for a representative tuned pathway-vector run. Cue-A and cue-B features segregate cleanly into separate latent pathways.

supports the structured-feedback extension studied here. To our knowledge, prior work has not tied exact path gains, direct gradient-fidelity diagnostics, and shunting inhibition into one causal account of when local dendritic learning works.

6 Discussion

We started from exact gradients for a conductance-based dendritic tree and asked when a somatic broadcast can serve as a useful proxy for compartment-specific errors. In the models studied here, the answer depends strongly on the inhibitory regime. Shunting changes the conductance denominator, stabilizes effective local sensitivities, improves agreement between local and backprop gradients, and yields better local learning than an additive control, especially in noisy or inhibition-dependent settings. An exploratory morphology $\times$ inhibition sweep (Appendix Fig. S10) suggests that tree shape shifts this operating regime: shallow trees peak at lower inhibition, while deeper trees retain their largest shunting advantage at modest inhibition. Explicit normalization recovers part of this benefit but not all of it, suggesting that shunting changes more than overall scale alone. This is therefore a mechanism paper, not a benchmark paper: the main contribution is to explain when simple broadcast-based local learning works in conductance dendritic networks.

Limitations. Our best accuracy (91.4% MNIST, 81.1% Fashion-MNIST) remains below several recent local-learning methods that operate in abstract compartments or standard activation spaces [36–39]. This reflects the constraints of conductance-based voltage space: bounded voltages, positive conductances, and a denominator-heavy computation graph. In the conservative flattened-CIFAR-10 sweep in Appendix Fig. S8, shunting local learning reaches 21.2% versus 40.1% backprop, indicating that the gradient-fidelity mechanism alone is insufficient for harder tasks without architectural advances such as convolutional encoders. Naive local broadcasts remain much weaker than per-soma on MNIST, and the cue-routing benchmark requires a still more structured pathway-vector signal (Sec. 4.7); our claims are therefore specific to the broadcast family studied here, not arbitrary local mismatch heuristics. Per-soma broadcast is low-bandwidth but still neuron-global because it assumes access to the full somatic error vector. Depth remains challenging: local learning degrades more than backprop as trees deepen (Appendix Fig. S6A), and at depth 4 shunting local learning is still roughly 33 percentage points below backprop. Shunting networks are also *not* gain-invariant in practice: exploratory global gain-shift pilots show sharp degradation because leak conductance breaks exact gain invariance. Finally, standard FA is incompatible with dendritic layer geometry (Appendix Fig. S7), and the weight-statistics analysis in Appendix G is descriptive rather than central evidence.

Broader relevance. The results are relevant to neuroscience, machine learning, and neuromorphic hardware. For neuroscience, they suggest a connection between divisive normalization and local credit fidelity. For machine learning, they show that conductance-based inductive biases can shape gradient geometry in ways that benefit local learning. For neuromorphic implementations, the update rules remain strictly local and parallelizable.

Testable predictions. Our framework predicts that (1) stronger perisomatic inhibition yields more precise synaptic plasticity; (2) GABA_A blockade selectively impairs multi-layer credit tasks; (3) the dendritic–somatic voltage correlation (ρ_n) increases during learning; and (4) learned excitatory weight distributions follow log-normal statistics whose width depends systematically on inhibitory conductance (Appendix G).

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Dataset	Rule	Top-10 valid	Top-10 test
MNIST	3F	0.611	0.622
MNIST	4F	0.620	0.628
MNIST	5F	0.912	0.916
Context gating	3F	0.398	0.396
Context gating	4F	0.411	0.411
Context gating	5F	0.807	0.789

Table S1: **Rule-family ranking.** Top-10 mean across completed local-competence sweeps.

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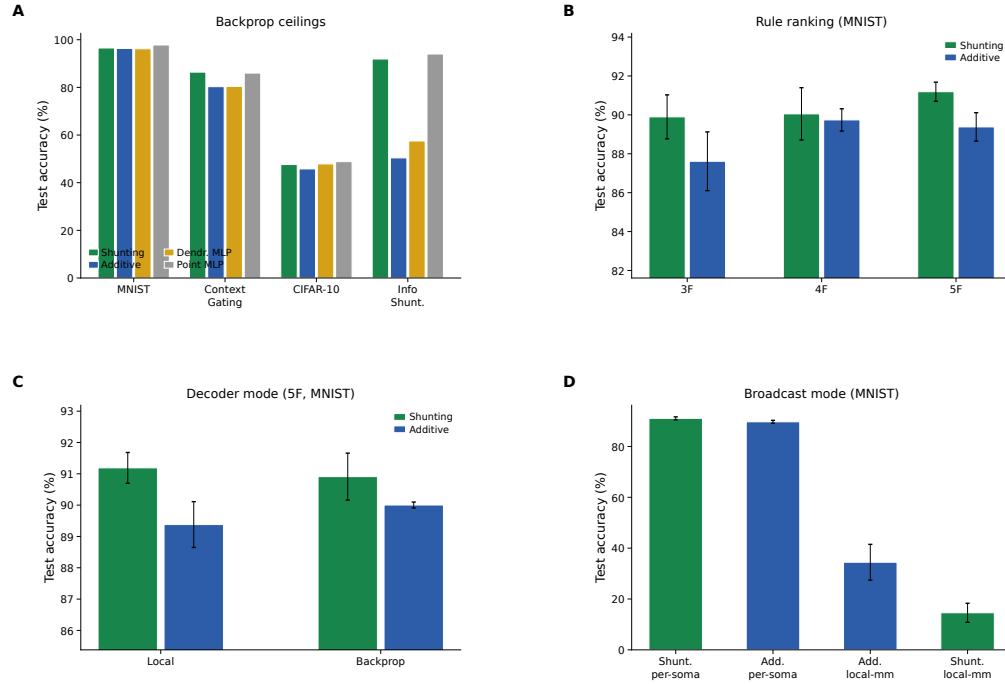


Figure S1: **Capacity calibration (supplementary).** (A) Phase 1 backprop ceilings across all architectures and datasets. (B) Rule-family ranking: 5F consistently outperforms 4F and 3F. (C) Decoder mode comparison (local vs. backprop decoder, 5F MNIST). (D) Broadcast mode comparison: per-soma is required for strong performance; local-mismatch fails.

408	A	Supplementary Results
409		Rule-Family Ranking
410		Which Components Carry?
411		Broadcast Mode Comparison and Local-Mismatch Recheck
412		Phase 1 Capacity Ceilings
413		Local Competence and Regime Dependence
414		Ablation Results
415		Extended Gradient Analysis and N_I Sweep Detail
416		Verification and Reproducibility
417		Additional Stress Tests
418		FA/DFA Baseline Comparison
419		CIFAR-10 Results

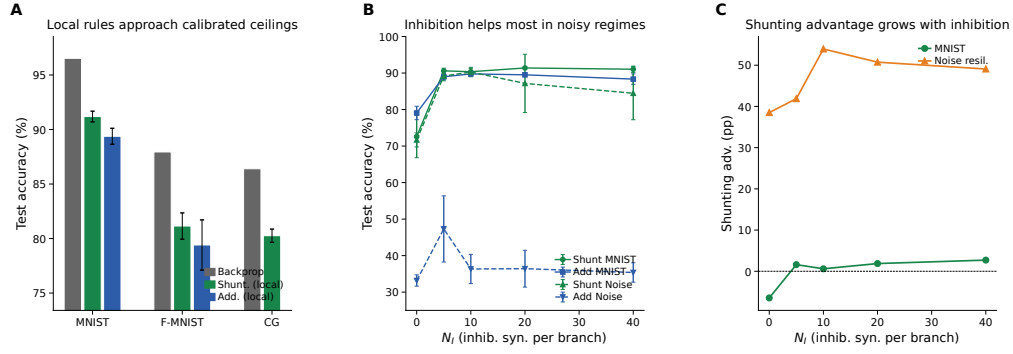


Figure S2: Local competence and regime dependence. (A) Backprop ceiling (gray) vs. best local rule (SF per-soma) for shunting (green) and additive (blue) cores on MNIST, Fashion-MNIST, and context gating. (B) N_I dose-response: test accuracy vs. inhibitory synapses per branch (N_I) for shunting and additive cores on MNIST (solid) and noise-resilience (dashed). Shunting needs $N_I \geq 5$ to reach high accuracy; additive learning fails on noise-resilience regardless of N_I . (C) Shunting advantage (Δ , in percentage points) vs. N_I . The shunting advantage grows with inhibitory synapses per branch and is larger on the noise-resilience task than on MNIST.

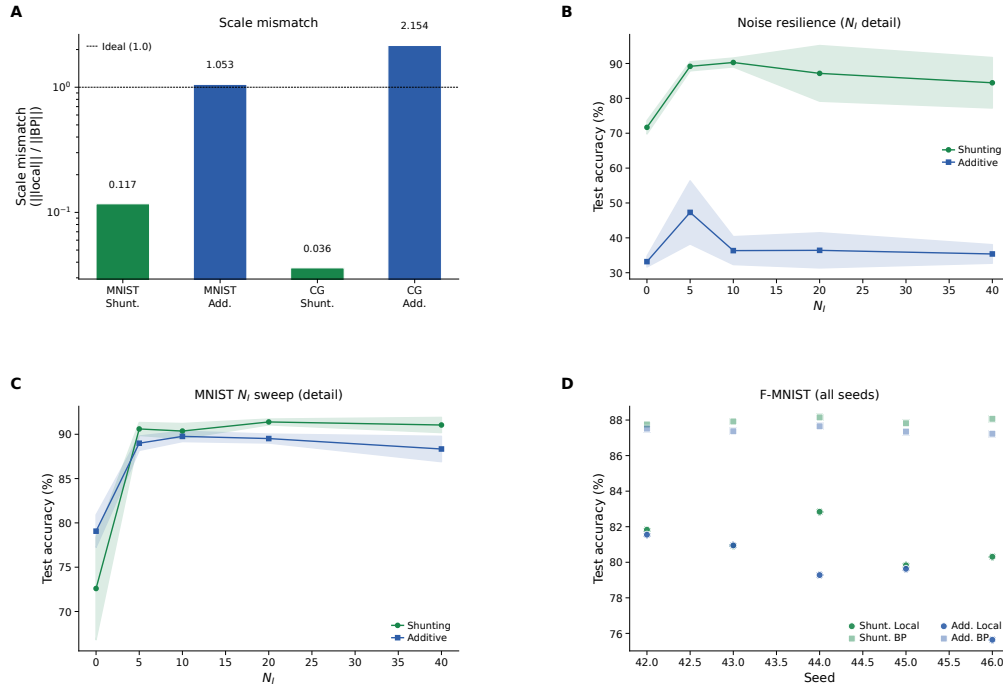


Figure S3: Extended gradient and N_I analysis (supplementary). (A) Scale mismatch bars: shunting achieves near-ideal scale (0.117); additive exhibits order-of-magnitude distortion (> 1.0). (B) Noise-resilience N_I dose-response with error bands (± 1 s.d.). (C) MNIST N_I dose-response detail with error bands. (D) Fashion-MNIST individual seeds for all conditions, showing consistency across runs.

Component	Representative effect	Interpretation
Shunting architecture	Path-gain CV stays at about 0.21–0.27 under shunting vs. 0.53–1.08 under additive; on noise resilience, per-soma local learning rises from roughly 0.33–0.47 (additive) to 0.72–0.90 (shunting).	This is the main architectural effect in the paper: shunting concentrates path gains and yields a more faithful low-bandwidth credit signal.
Broadcast quality	On MNIST (5F), shunting per-soma reaches 0.912 ± 0.005 , while local-mismatch remains at 0.146 ± 0.046 .	Correctly matching the broadcast field to the somatic error matters far more than arbitrary local mismatch heuristics.
Transported oracle	On noise resilience, additive local learning rises from 33–47% under per-soma to 75.6–98.5% under transported broadcast; shunting rises from 71.7–90.3% to 98.5–99.3%.	The oracle is the strongest causal evidence that broadcast fidelity, not just architecture, is the dominant bottleneck.
Rule hierarchy	In the competence sweeps, MNIST improves from 0.622 (3F) and 0.628 (4F) to 0.916 (5F); context gating improves from 0.396/0.411 to 0.789.	5F is the strongest practical rule in this paper, though its extra factor remains heuristic rather than theorem-derived.
5F stability	Tightening the ϕ clamp to $[0.5, 2.0]$ changes MNIST by only about +0.4 pp; varying EMA from 0.05 to 0.20 changes it by less than 0.2 pp.	The practical gain from 5F is not especially brittle to its main tuning knobs.
Structured feedback	On learned cue routing, tuned shunting per-soma reaches $88.2 \pm 13.9\%$, while tuned pathway-vector feedback reaches $99.1 \pm 0.3\%$.	Structured feedback matters when different branches must be credited differently; it is not a generic replacement for per-soma broadcast on ordinary tasks.
Random low-rank channels	On noise resilience, random low-rank broadcast improves from 0.456 at $K=1$ to 0.606 at $K=4$, but remains far below the stronger per-soma and transported-broadcast baselines; on cue routing, raising random rank from $K=1$ to $K=2$ does not match PV-LocalCA.	Dimensionality alone is not enough: pathway structure matters, not just additional random broadcast channels.
Morphology regime	In the exploratory noise-resilience regime map (Appendix Fig. S10), depth-2 trees show their largest shunting gains near $N_I = 0$, whereas depth-3 trees retain their strongest gains around $N_I = 5$.	Morphology appears to shift the inhibitory operating regime in which shunting is most useful, suggesting that conductance normalization and tree geometry interact rather than contributing independently.

Table S2: **Which components carry the observed gains?** A compact summary of the empirical division of labor across architecture, broadcast design, and LocalCA rule variants.

Core	Broadcast	Decoder	Test (mean $\pm$ std)
Shunting	per-soma	local	0.912 $\pm$ 0.005
Shunting	per-soma	backprop	0.909 $\pm$ 0.008
Shunting	local-mismatch	local	0.146 $\pm$ 0.046
Shunting	local-mismatch	backprop	0.146 $\pm$ 0.037
Additive	per-soma	local	0.894 $\pm$ 0.007
Additive	per-soma	backprop	0.900 $\pm$ 0.001
Additive	local-mismatch	local	0.342 $\pm$ 0.058
Additive	local-mismatch	backprop	0.348 $\pm$ 0.095

Table S3: **Local-mismatch recheck (MNIST, 5F)**. Per-soma is consistently strong; local-mismatch remains much weaker.

Dataset	BP ceiling	Best local (5F)	Gap
MNIST	0.965	0.914 $\pm$ 0.003	5.1%
Fashion-MNIST	0.879	0.811 $\pm$ 0.012	6.8%
Context gating	0.864	0.803 $\pm$ 0.006	6.1%

Table S4: **Local competence**. Backprop ceilings are matched-capacity selection estimates: MNIST and context gating use the best phase-1 standard-training setting from a 2-seed capacity-calibration sweep, whereas Fashion-MNIST uses a dedicated 5-seed competence sweep. Local values are 5F with per-soma broadcast on shunting dendritic cores. Context gating additionally uses an HSIC auxiliary objective (weight 0.01; Appendix E). Error bars on local values: ± 1 s.d. across 5 seeds.

Condition	Metric	Value
Decoder: local vs backprop vs frozen (MNIST, sandbox)	test acc	0.379 vs 0.379 vs 0.176
Shunting vs additive, matched (MNIST, sandbox)	Δ test	+0.210
Per-soma, path on vs off (synthetic task, sandbox)	test / MI(E,I;C)	-0.003 / +0.017

Table S5: **Mechanistic ablations**. Results from controlled small-network sandbox experiments using minimal architectures to isolate individual factors. Sandbox runs use a small 2-layer dendritic network (branch factor [9]) trained for 30 epochs; they are not directly comparable to the headline capacity-calibrated results.

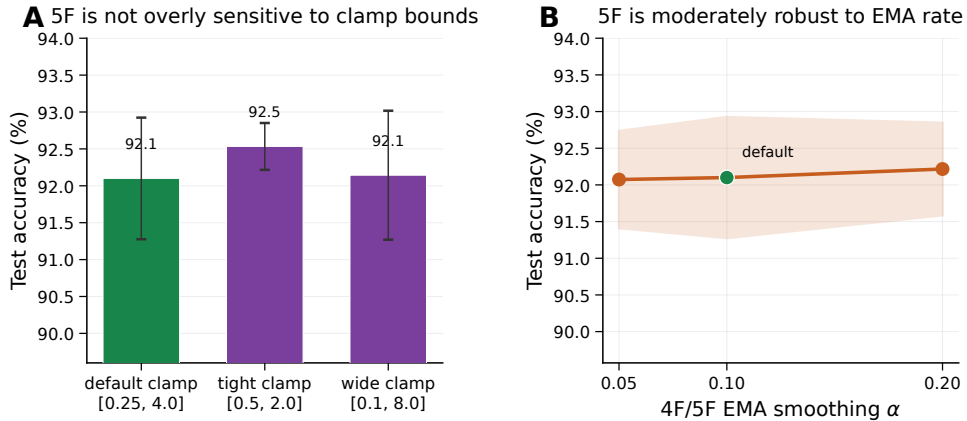


Figure S4: **5F sensitivity analysis (MNIST, shunting, per-soma broadcast)**. (A) Sensitivity to the ϕ_n clamp interval. The default clamp [0.25, 4.0] is already strong (92.1 $\pm$ 0.8%); tightening it to [0.5, 2.0] slightly improves performance (92.5 $\pm$ 0.3%), while widening it to [0.1, 8.0] is effectively neutral (92.1 $\pm$ 0.9%). (B) Sensitivity to the 4F/5F EMA rate used for online statistics. Varying α from 0.05 to 0.20 changes mean test accuracy by less than 0.2 pp. Together these results suggest that the 5F gain is not driven by a brittle single hyperparameter choice, even though the factor itself remains heuristic. Errors: ± 1 s.d. across 3 seeds.

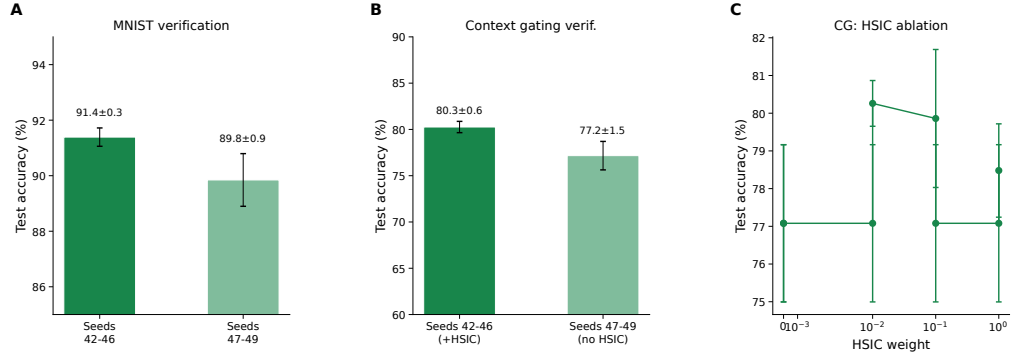


Figure S5: **Verification and reproducibility (supplementary).** (A) MNIST verification: main seeds (42–46) yield $91.4 \pm 0.3\%$; held-out seeds (47–49) yield $89.8 \pm 0.9\%$, confirming generalization. (B) Context gating verification: main seeds (+HSIC) $80.3 \pm 0.6\%$; held-out seeds (no HSIC) $77.2 \pm 1.5\%$. The roughly 3 pp gap reflects HSIC removal, not seed sensitivity. (C) HSIC weight ablation on context gating: moderate weights (0.01–0.1) perform best.

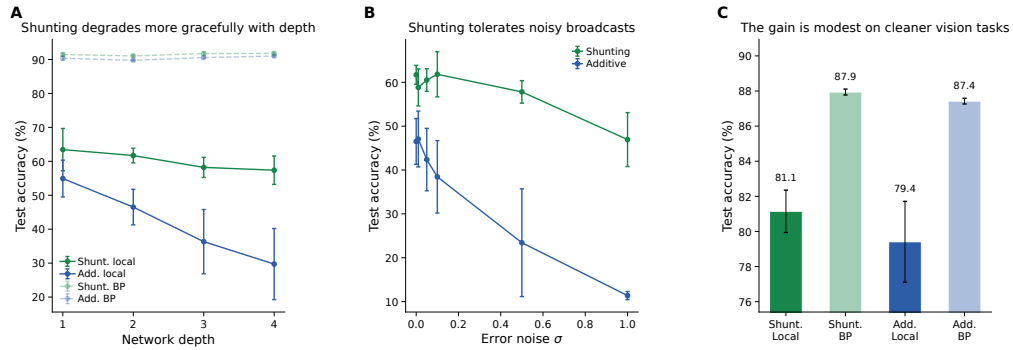


Figure S6: **Additional stress tests.** (A) Depth scaling: shunting local (green) degrades more gracefully from 63.5% to 57.4%; additive local (blue) drops from 54.9% to 29.7%. Backprop ceilings (dashed) remain stable. (B) Broadcast-noise robustness: shunting remains stable for $\sigma \leq 0.1$; additive degrades rapidly and reaches chance at $\sigma = 1$. (C) Fashion-MNIST: the shunting gain is modest on this cleaner benchmark (81.1% local vs. 87.9% backprop), consistent with the regime-dependent story in the main text.

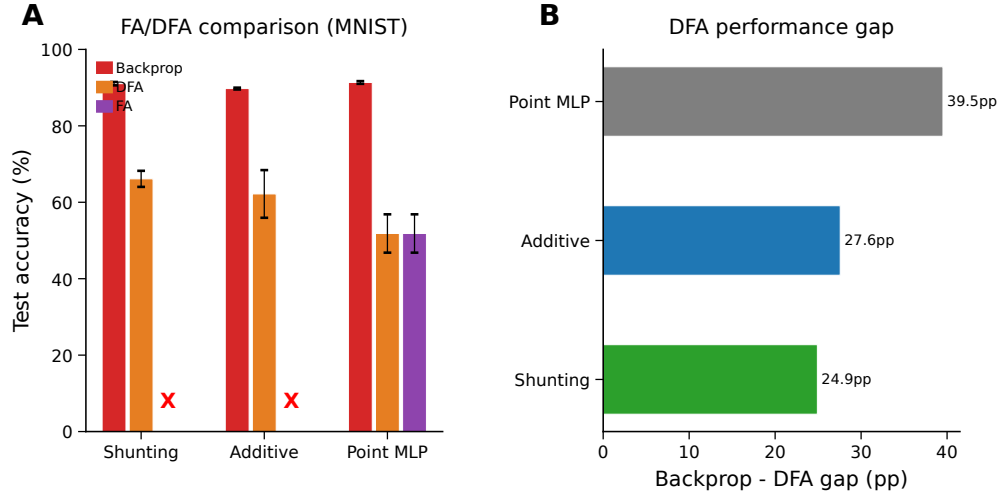


Figure S7: **Feedback alignment baselines (MNIST)**. (A) Grouped comparison: standard backprop (gray), DFA (orange), and FA (purple). Red X marks indicate FA failure on dendritic architectures (dimensional incompatibility between random feedback matrices and block-structured dendritic layers). DFA achieves 66.1% on shunting vs. 62.2% additive vs. 51.8% point MLP. (B) Backprop–DFA gap: dendritic architectures (24.9–27.6 pp) show smaller gaps than point MLPs (39.5 pp), suggesting conductance-based architecture is partially compatible with random feedback.

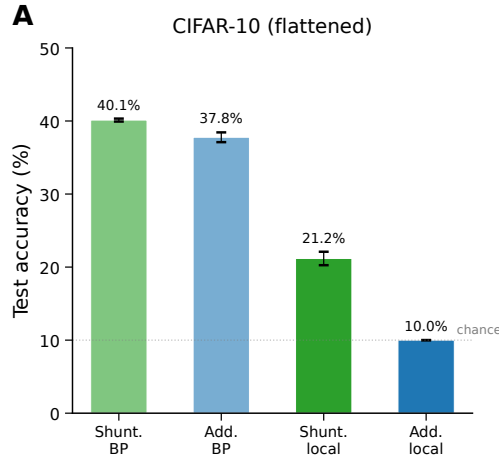


Figure S8: **Conservative flattened CIFAR-10 reference sweep**. In this appendix stress test, shunting backprop reaches 40.1%, additive backprop 37.8%, shunting local 21.2%, and additive local 10.0% (chance). The backprop–local gap (about 19 pp for shunting) is larger than on MNIST, reflecting the increased difficulty of propagating credit through conductance-based layers for complex visual features. This panel should be read as a conservative flattened-image benchmark rather than as the tuned CIFAR ceiling for the architecture family.

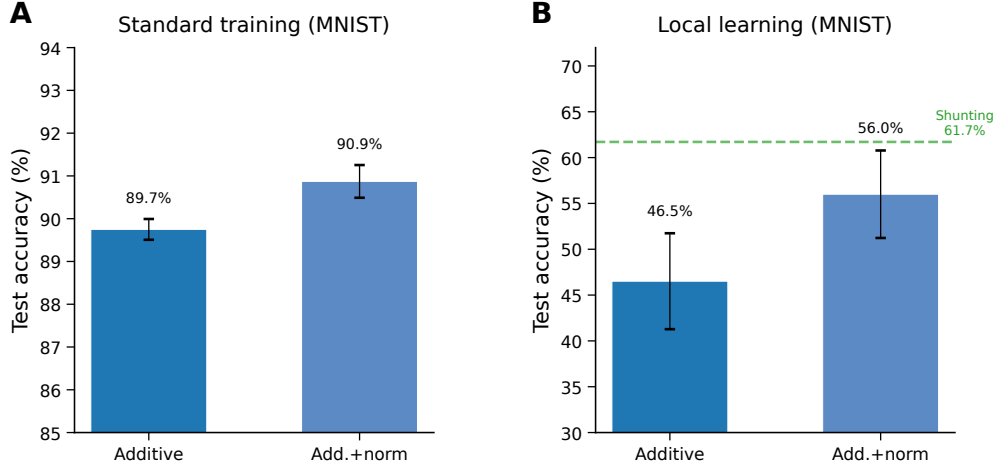


Figure S9: **Additive + normalization control (MNIST)**. (A) Standard backprop: normalization provides a small boost (89.7% $\rightarrow$ 90.9%). (B) Local learning: normalization improves additive from 46.5% to 56.0% (+9.5 pp), partially closing the gap to shunting (61.7%, dashed green). The remaining gap (−5.7 pp) indicates that shunting provides benefits beyond simple voltage normalization.

Broadcast rank K	Noise resilience	Cue routing
1	0.456 ± 0.111	0.877 ± 0.084
2	0.547 ± 0.097	0.839 ± 0.106
4	0.606 ± 0.044	—
8	0.546 ± 0.100	—
10	0.563 ± 0.104	—

Table S6: **Random low-rank broadcast does not replace structure**. On noise-resilience, a small number of random broadcast channels helps somewhat, with a weak optimum near $K=4$, but remains far below the stronger per-soma and transported-broadcast baselines in the main text. On cue routing, increasing unstructured rank from $K=1$ to $K=2$ does not recover the pathway-aware gains of PV-LocalCA.

420 Additive Normalization Control

421 Morphology $\times$ Inhibition Regime Map

422 Random Low-Rank Broadcast Channels

423 B Implementation Details

424 Biological Plausibility Assumptions

Each synapse has access to: presynaptic activity x_j (local), compartment voltage V_n (local membrane potential), reversal potential E_j (fixed by receptor type), and total input resistance $R_n^{\text{tot}} = 1/g_n^{\text{tot}}$ (computable from local conductances). The 4F modulator ρ_n requires online estimation of voltage covariance (biologically plausible via slow calcium signals); the 5F factor ϕ_n requires a linear regression proxy (implementable via eligibility traces). The only non-local quantity is the broadcast error e_n , which requires a top-down or neuromodulatory signal from the soma to dendritic compartments. We assume per-neuron resolution (one error per output neuron) for the per-soma broadcast mode.

432 Dendritic Structure in Code

We use the morphology notation $[b_1, b_2, \dots, b_D]$ for rooted dendritic trees, where each factor gives the fan-in from one dendritic stage to the next on the path to the soma. For example, $[3, 3]$ means that each soma aggregates 3 proximal branches and each proximal branch aggregates 3 distal branches, for

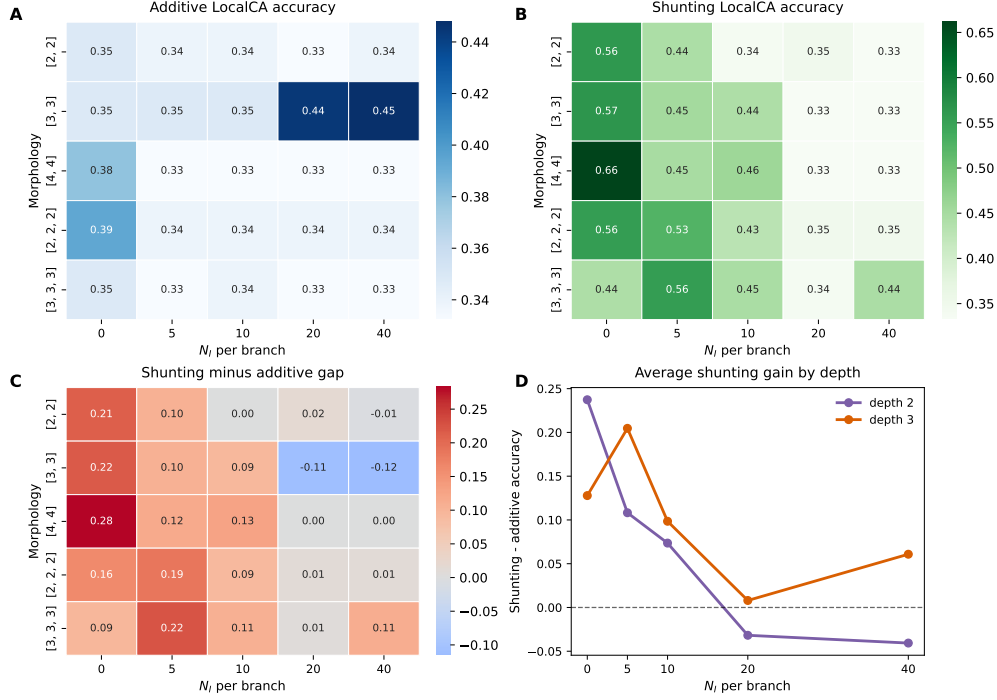


Figure S10: **Exploratory morphology × inhibition regime map on noise resilience.** (A) Additive LocalCA remains comparatively flat across morphologies and inhibitory synapse counts, with only isolated high-variance pockets. (B) Shunting LocalCA shows a clear morphology-dependent regime: the strongest performance appears at low or moderate inhibition, and the best N_I shifts with tree shape. (C) The shunting-minus-additive gap is largest for wider two-level trees at low inhibition and for deeper trees at modest inhibition, then weakens or vanishes at high inhibition. (D) Averaging by depth highlights the same shift: depth-2 trees peak near $N_I = 0$, whereas depth-3 trees retain their largest shunting advantage around $N_I = 5$. We treat this figure as exploratory support for the claim that morphology changes the inhibitory operating regime in which shunting is most useful; a fuller mechanistic account would require matched theory diagnostics for each cell in the grid.

Quantity	Symbol	Convention
Voltage	V	Normalized to $[-1, 1]$
Conductances	$g^{\text{syn}}, g^{\text{den}}$	Nonneg. via softplus
Leak conductance	g^{leak}	Set to 1
Input resistance	R^{tot}	≤ 1

Table S7: Units and normalization.

9 distal leaves per soma; [3, 3, 3] adds a third dendritic stage with 27 leaves per soma. In code, each
excitatory unit is instantiated as a `DendriNet`, which is a stack of `DendriticBranchLayers`. Each
branch layer contains an excitatory synapse bank, an optional inhibitory synapse bank, and a branch-to-
parent aggregation map. In the main feedforward models studied here, `somatic_synapses=false`,
so synaptic inputs terminate on dendritic branches rather than directly on the soma. Likewise,
`input_mode=1` together with nonzero `ie_synapses_per_branch_per_layer` means that inhibition
enters each dendritic branch through explicit inhibitory synapses even when we do not instantiate
a separate inhibitory neuron population (`inhibitory_layer_sizes=[]`). The additive and shunt-
ing cores are therefore architecture-matched at the tree level; they differ in the branch-level voltage
integration rule rather than in whether a dendritic tree exists.

Category	Implemented modes	Role in this paper
Core models	point MLP; dendritic MLP; dendritic additive; dendritic shunting; pathway-routed dendritic cores	Main comparisons use dendritic additive vs. dendritic shunting; routed cores are used for cue integration.
Training strategies	standard backprop; LocalCA; DFA; FA; transported-broadcast oracle	Main text uses standard and LocalCA; DFA/FA remain appendix baselines; path transport is an oracle upper bound.
Broadcast modes	scalar; per-soma; local mismatch; low-rank; path transport; pathway vector	Main text centers on per-soma and pathway-vector; scalar and local-mismatch are supporting controls; path transport is diagnostic only.
Recent dendritic extensions	path propagation; router-derived branch roles; pathway-vector feedback; inhibitory homeostasis	These were added to test morphology-aware credit transport and structured branch-specific plasticity beyond the original 3F/4F/5F family.

Table S8: Implemented model families and LocalCA operating modes. Only a subset is used for the main claims; the remaining modes serve as controls, ablations, or structured extensions.

Setting	Encoder	E layers	Tree	Synapses / branch	Seeds	Notes
Gradient fidelity / path-gain / noise-resilience mechanism	identity	[128]	[3, 3]	$N_E=40, N_I \in \{0, 5, 10, 20, 40\}$	5 fidelity, 3 oracle	Main mechanistic sweeps in Figs. 2 and 3.
MNIST / context-gating local competence	identity	[128]	[3, 3]	$N_E=40, N_I=20$	5	Best local 5F per-soma runs used in Table S4.
Fashion-MNIST competence	identity	[128]	[3, 3]	$N_E=40, N_I=20$	5	Dedicated five-seed standard-vs-local competence sweep.
Cue-routing PV-LocalCA	pathway router	[64]	[2]	$N_E=10, N_I=4$	5	Learned router with two pathway groups and rank-2 structured feedback.
Morphology×inhibition appendix map	identity	[128, 128]	[2, 2], [3, 3], [4, 4], [2, 2, 2], [3, 3, 3]	$N_E=40, N_I \in \{0, 5, 10, 20, 40\}$	3	Exploratory appendix sweep only.

Table S9: **Representative architectures used in the manuscript.** All dendritic rows use explicit branch-level excitatory and inhibitory synapse banks with `somatic_synapses=false` and nonnegative conductances. The tree column gives the dendritic morphology per excitatory neuron, while the synapse counts report excitatory and inhibitory synapses per branch.

446 Units and Parameterization

447 Decoder Update Modes

448 W_{dec} maps $V_L \rightarrow \hat{y}$. Modes: **backprop** ($\nabla_W L$ via autograd), **local** ($\Delta W = \eta \langle \delta_0 V_L^\top \rangle_B$), **frozen**
449 ($\Delta W = 0$).

Extension	Meaning	Compatible with	Typical use here
Per-soma broadcast	Somatic error vector shared across compartments	3F, 4F, 5F	Default main-text LocalCA setting
Low-rank broadcast	Small number of broadcast channels instead of a scalar or full field	3F, 4F, 5F	Intermediate control when rank-1 is too restrictive
Path transport	Recursive tree-transported error using path gains	3F, 4F, 5F	Oracle upper-bound broadcast
Path propagation	Approximate path-gain attenuation applied to broadcast	3F, 4F, 5F	Morphology-aware modifier for deeper trees
Pathway vector	Structured branch/pathway-specific broadcast field	3F, 4F, 5F	Used mainly in PV-LocalCA on routed tasks
Branch-role rules	Scales updates by router- or pathway-derived branch specialization	3F, 4F, 5F	Used in the structured pathway extension
Inhibitory homeostasis	Auxiliary local inhibitory balancing term	3F, 4F, 5F	Optional stabilizer in routed or shunting settings

Table S10: Compatibility of the newer LocalCA extensions with the base 3F/4F/5F family. These extensions mostly modify the error field or branch scaling rather than defining a new “6F” or “7F” rule.

Model and Mode Inventory

Representative Manuscript Architectures

LocalCA Extension Compatibility

Seed Protocol and Claim Status

The seed depth is not uniform across every sweep, and we therefore separate headline evidence from exploratory screening. The main mechanistic gradient-fidelity sweep uses 5 seeds per condition, the cue-routing seed-repeat and tuning sweeps use 5 seeds per condition, the dedicated Fashion-MNIST competence sweep uses 5 seeds per condition, the path-transport oracle sweep uses 3 seeds per condition, and the 5F sensitivity sweep uses 3 seeds per condition. By contrast, early architecture-selection or stress-test screens with only 1–2 seeds are treated as exploratory and are either omitted from the main claims or explicitly framed as appendix-only pilots. For transparency, the BP ceilings in Table S4 for MNIST and context gating come from a matched-capacity phase-1 calibration sweep with 2 seeds, whereas the corresponding local runs use 5 seeds and the Fashion-MNIST standard/local comparison uses a dedicated 5-seed sweep. This asymmetry affects the confidence we should place in the ceiling estimates, not the mechanistic sweeps that anchor the paper’s main claims.

Compute Resources

All experiments were run on an internal GPU cluster. Most MNIST, Fashion-MNIST, and synthetic-task runs used a single NVIDIA A100 or H100 GPU and finished in minutes to tens of minutes per seed; the flattened CIFAR-10 runs required longer single-GPU jobs but still did not use distributed or multi-GPU training. Reported headline metrics are aggregated over 3–5 seeds as stated in the text; the broader exploratory sweeps used to tune and diagnose the method required additional compute beyond the final reported runs.

Code, Data, and Asset Availability

MNIST, Fashion-MNIST, CIFAR-10, and MICrONS are public datasets or public benchmark assets cited in the paper; we use them under their standard published access conditions and cite their original sources in the bibliography. The present anonymous submission does not include a public code URL in order to preserve double-blind review, but the paper specifies the model family, training modes, key

Rule	Factors	Cost	Best regime
3F	$x, (E - V), e$	$\mathcal{O}(1)$	Baseline
4F	$3F + \rho$	$\mathcal{O}(1)$	Improved conditioning
5F	$4F + \phi$	$\mathcal{O}(d_n)$	Strongest overall
5F+VH	5F + local voltage homeostasis	$\mathcal{O}(d_n)$	Stabilized LocalCA / warm-up
PV-LocalCA	3F + structured e_n + role alignment	$\mathcal{O}(r_n)$	Pathway discovery

Table S11: Variant taxonomy. ‘5F+VH’ denotes the same base 5F rule with an additional strictly local voltage-centering auxiliary that can also be scheduled through multi-stage LocalCA warm-up; the codebase also supports this as a shorthand ‘5f_vh’ alias in configuration files.

hyperparameters, and representative configurations, and we plan to release code and configuration files after review. The paper does not release a new dataset or a stand-alone pretrained model asset.

Algorithm

Algorithm 1 Local Credit Assignment

```

1: Input: Model, batch  $(x, y)$ , config  $\mathcal{C}$ 
2: Forward pass; loss  $L$ , output error  $\delta^y$ 
3: Somatic error  $\delta_0 = W_{\text{dec}}^\top \delta^y$ 
4: for each layer  $n$  (reverse) do
5:    $e_n = \text{broadcast}(\delta_0, \mathcal{C})$ ; optionally gate by pathway roles
6:   Compute  $\rho_n, \phi_n$  (EMA estimators, if enabled)
7:   Apply the configured local update (Eq. 6 or 7)
8: end for
9: Clip gradients; optimizer step

```

C Theoretical Details

Variant Taxonomy

Random-Broadcast Alignment

Proposition 3 (Supportive random-broadcast alignment). *Let the broadcast matrix B_n have i.i.d. zero-mean entries with $\mathbb{E}[B_n^\top B_n] = \alpha I$. Then the expected cosine between the resulting local gradient and the exact gradient is positive, $\mathbb{E}[\cos \angle(g^{\text{local}}, g^{\text{exact}})] \geq c_n > 0$, by an argument analogous to feedback alignment [9]. The constant c_n depends on how strongly local eligibility factors correlate with the exact path gain in Eq. (4); shunting improves this correlation by narrowing the spread of intermediate sensitivities.*

We treat this argument as supportive rather than central. The main theoretical object in the paper is still the exact compartment error $\partial L / \partial V_n = \alpha_n \delta_0$, and the main empirical question is how faithfully different broadcast fields approximate that quantity.

Biological Analogs

D Morphology-Aware Extensions

Path-integrated propagation. Modulate broadcast error by $\pi_n = \pi_{n-1} \cdot R_{n-1}^{\text{tot}} \cdot \bar{g}_{n-1}^{\text{den}}$, approximating depth attenuation from Eq. 4.

Depth modulation. Per-branch scaling $\rho_j = \rho_{\text{base}} / (d_j + \alpha)$, mirroring cable attenuation.

Dendritic normalization. $\Delta g_j^{\text{den}} \leftarrow \Delta g_j^{\text{den}} / (\sum_k g_k^{\text{den}} + \varepsilon)$, analogous to homeostatic scaling [23].

Component	Analog	Interpretation
R_n^{tot}	Input resistance	Sensitivity modulation
$(E_j - V_n)$	Synaptic driving force	Local gradient factor
Shunting	Divisive normalization	$\partial V / \partial g_I \propto -V$
ρ_n	Layer relevance	Output correlation
ϕ_n	Signal propagation	Conditional predictability
Pathway vector	Branch-specific top-down drive	Structured teaching signal
Role alignment	Pathway-selective plasticity	Branch specialization

Table S12: Biological analogs.

Pathway-vector broadcast. For tasks with latent pathway structure, the broadcast becomes a low-dimensional vector over router-inferred pathway roles. This vector is gated by local pathway activity and propagated through the block-structured tree so that earlier branches receive feedback aligned with their downstream descendants.

Router-derived branch roles. Each branch receives a role profile inferred from router assignments and incoming block weights rather than from a hand-coded apical/basal label. Plasticity is then modulated by branch selectivity and a synapse-specific role-alignment factor, emphasizing pathway-consistent updates without imposing a heuristic branch taxonomy.

E HSIC Auxiliary Objectives

Following [18], we use kernel-based HSIC objectives. Self-decorrelation: $\mathcal{L}^{\text{self}} = B^{-2} \text{tr}(\mathbf{K}_Z \mathbf{H} \mathbf{K}_Z \mathbf{H})$. Target-correlation: $\mathcal{L}^{\text{target}} = -B^{-2} \text{tr}(\mathbf{K}_Z \mathbf{H} \mathbf{K}_Y \mathbf{H})$. Moderate weights (0.01–0.1) help on context gating; negligible on MNIST. Online statistics (ρ_n, ϕ_n) use Welford’s algorithm [22].

F Online Variant with Eligibility Traces

Continuous-time eligibility: $\tau_e \dot{e}_j^{\text{syn}} = -e_j^{\text{syn}} + x_j(E_j - V_n)R_n^{\text{tot}}$. Update: $\Delta g_j^{\text{syn}} \propto \int e_j^{\text{syn}}(t)m_n(t) dt$ [19, 20].

G Weight Statistics and Biological References

As a secondary analysis, we examine whether the conductance-based update naturally induces characteristic weight statistics. Because synaptic updates scale with input resistance $R_n^{\text{tot}} = 1/g_n^{\text{tot}}$, the voltage equation implies multiplicative weight dynamics, creating a regime analogous to exponentiated gradient descent [43]. Such multiplicative dynamics suggest log-normal weight distributions in trained networks [41, 42]. We compare the resulting distributions against two biological references: electrophysiological data from Song et al. [41] (rat V1 L5 pyramidal, $\sigma = 0.94$) and the MICrONS mm³ connectome [44] (mouse V1 synapse sizes).

Model weight distributions are log-normal. We fit log-normal distributions to the active (top- K masked) excitatory synaptic weights after training on MNIST and Fashion-MNIST. Table S13 shows the fitted σ (standard deviation of $\log w$) for each condition. Shunting consistently produces narrower distributions than additive integration, and backpropagation produces narrower distributions than local learning. These two factors combine approximately additively: BP+Shunting $\rightarrow$ narrowest, Local+Additive $\rightarrow$ broadest.

Shunting is closest to the biological references considered here. BP+Shunting produces $\sigma = 0.94$ on MNIST, very close to the Song et al. reference ($\Delta = 0.008$). Local+Shunting produces $\sigma = 1.13$ – 1.22 , close to MICrONS E $\rightarrow$ E synapse sizes ($\sigma = 1.14$; $\Delta \leq 0.08$). By contrast, Local+Additive produces $\sigma = 1.43$ – 1.69 , broader than any biological reference considered. This pattern is consistent across datasets (MNIST and Fashion-MNIST), suggesting it reflects structural properties of the learning rule rather than task-specific features.

Condition	MNIST		Fashion-MNIST	
	Exc σ	Inh σ	Exc σ	Inh σ
BP + Shunting	0.94±0.01	0.87±0.01	0.92±0.01	0.84±0.01
BP + Additive	1.30±0.01	1.22±0.01	1.16±0.01	1.05±0.01
Local + Shunting	1.22±0.01	1.71±0.02	1.13±0.06	1.36±0.06
Local + Additive	1.69±0.16	1.57±0.07	1.43±0.16	1.11±0.12
Song et al. 2005	$\sigma = 0.94$ (rat V1 L5, EPSP amplitude)			
MICrONS E→E	$\sigma = 1.14$ (mouse V1, synapse size, $n=3.9M$)			
MICrONS I→E	$\sigma = 0.92$ (mouse V1, synapse size, $n=3.5M$)			

Table S13: **Weight distribution width.** Log-normal σ of excitatory and inhibitory synaptic weights (mean±std across 5 seeds). Biological references are included only as descriptive anchors. With biologically constrained E/I ratios (4:1), shunting self-normalization produces weight heterogeneity in the range spanned by those references, whereas additive integration is broader and less stable.

The match depends on inhibitory conductance. We test sensitivity to the E/I synapse ratio using a sweep over $N_I \in \{0, 2, 5, 10, 20, 30, 40\}$ inhibitory synapses per branch (Appendix Fig. S11). Without inhibition ($N_I=0$), all conditions produce $\sigma > 2.0$, far from biology. With biologically constrained ratios ($N_I=10-20$, corresponding to E/I ratios of 4:1 to 2:1; [46]), shunting networks enter the biological range ($\sigma \approx 0.9-1.2$), while the effect of N_I on additive networks is less systematic. The weight distribution shape is also robust to dendritic depth (branch factors [9] through [3,3,3,3]; see Appendix Fig. S11A): BP+Shunting converges toward Song et al. with increasing depth, while Local+Shunting remains near MICrONS E→E across depths.

H Weight Distribution Sensitivity Analysis

MICrONS connectome analysis. We extracted synapse size distributions from the MICrONS mm³ condensed connectome (v1181; 9.0M connections, 71.9K neurons; 89% excitatory). Splitting by pre/post cell type: E→E connections ($n=3.85M$) show $\sigma_{\text{size}}=1.14$ and CV=1.01; I→E ($n=3.53M$) show $\sigma=0.92$ and CV=1.03; E→I ($n=1.11M$) show $\sigma=0.83$; I→I ($n=0.51M$) show $\sigma=0.85$. Inhibitory connections are more frequently multi-synaptic (I→E: 37.5% multi-synapse; E→E: 11.5%), and inhibitory synapses are on average smaller (median 3532 vs. 4676 for E→E).

Sensitivity to ee_synapses. Using the E/I grid sweep ($N_E \in \{20, 40, 80\} \times N_I \in \{0, 2, 5, 10, 20, 30, 40\}$; 168 models), excitatory σ increases modestly with N_E in both architectures (+0.4–0.6 from $N_E=20$ to 80), but the shunting–additive ordering is preserved at all N_E values. The dominant driver of σ is N_I : going from $N_I=0$ to $N_I=10$ reduces shunting σ by approximately 1.0–1.5 units, while the effect on additive is smaller and less systematic.

Sensitivity to depth. From the depth-scaling sweep ([9], [3,3], [3,3,3], [3,3,3,3]; 80 models, 5 seeds each), BP+Shunting excitatory σ decreases with depth ($1.06 \rightarrow 0.90$), converging toward Song et al. (0.94). Local+Shunting σ increases slightly ($1.18 \rightarrow 1.33$) but remains in the biological range at shallow depths. BP+Additive shows the strongest depth sensitivity ($1.38 \rightarrow 0.79$), while Local+Additive diverges with depth ($2.01 \rightarrow 2.44$). The morphology-aware modulators (path propagation, depth/centrality weighting) have negligible effect on σ .

Mechanistic interpretation. The shunting conductance $g_n^{\text{tot}} = g_n^{\text{leak}} + \sum g_n^{\text{exc}} + \sum g_n^{\text{inh}}$ appears in the update factor $R_n^{\text{tot}} = 1/g_n^{\text{tot}}$. As inhibitory conductance increases, g_n^{tot} grows and R_n^{tot} shrinks, compressing the multiplicative scale of weight updates. This is analogous to a learning rate that self-adjusts via the total conductance: strong synapses contribute more to g_n^{tot} , reducing their own future update magnitude. This acts as a natural form of weight-dependent regularization. The resulting equilibrium distribution is log-normal, consistent with Loewenstein et al. [45] (multiplicative spine dynamics) and Saxena et al. [43] (exponentiated gradient). Additive integration lacks this self-normalizing mechanism, producing broader distributions that fall outside the biological range.

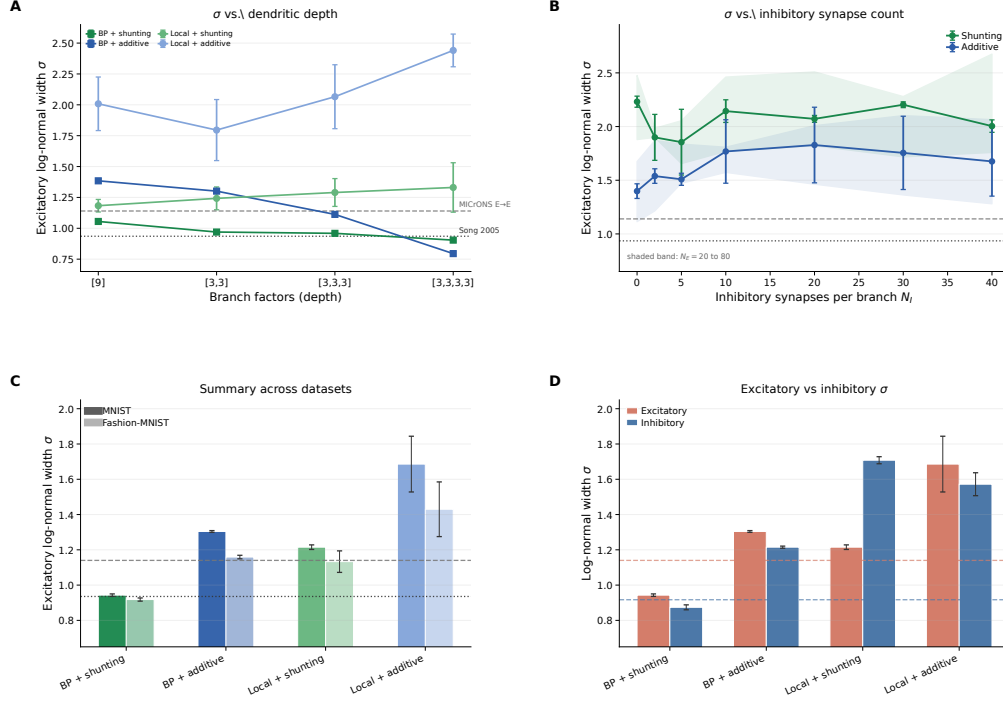


Figure S11: **Synaptic weight distribution sensitivity to architecture.** (A) Log-normal σ vs. dendritic depth (branch factors [9] to [3,3,3,3]). Both backprop and local shunting remain closer to the biological-width references than their additive counterparts across depth. (B) σ vs. inhibitory synapses per branch (N_I). Without inhibition ($N_I=0$), all conditions are overly broad; with biologically constrained $N_I=10$ –20, shunting enters the biological range. Shaded bands show the sweep over $N_E=20$ to 80. (C) Summary of MNIST and Fashion-MNIST results with biological references. The shunting ordering is consistent across datasets. (D) Excitatory vs. inhibitory weight σ on MNIST with MICrONS E→E and I→E reference bands.

569 I Synthetic Benchmark Descriptions

570 Each synthetic task is generated procedurally from a fixed random seed and uses fixed train/val/test
 571 splits for reproducibility. All tasks use 500-dimensional flattened inputs and 10-class outputs unless
 572 otherwise noted.

573 **Context gating.** Inputs are Gaussian mixtures over 10 classes, but the class-conditional means are
 574 linearly transformed by one of $C = 4$ context configurations. A one-hot context vector (appended to
 575 the input) indicates which configuration is active. Without the context, classes are ambiguous; with it,
 576 the optimal boundary shifts. This task tests whether credit assignment can exploit the context gating
 577 structure to modulate branch-specific representations appropriately. The HSIC auxiliary objective
 578 (weight 0.01) is used to decorrelate layer representations, adding roughly 3 pp under local learning.

579 **Noise resilience.** Inputs are MNIST features corrupted by structured channel noise: a random
 580 50×50 projection matrix adds correlated noise drawn from $\mathcal{N}(0, \sigma_{\text{task}}^2 I)$ with $\sigma_{\text{task}} = 1.5$. The
 581 noise projection is fixed across training; the network must filter structured interference to recover the
 582 underlying class signal. Credit assignment must remain coherent under the corrupted signal, making
 583 this task particularly sensitive to path-gain dispersion.

584 **Info shunting.** Inputs contain two additive streams: a low-frequency signal and a high-frequency
 585 distractor drawn with equal energy. A second context input (separate channel) indicates whether the
 586 relevant signal is low- or high-frequency on each trial. Correct classification requires the network to
 587 gate the distractor stream, a computation that naturally invites inhibitory modulation. Performance

at $N_I=0$ is notably below shunting performance (+24.8 pp), reflecting the task’s dependence on divisive normalization.

Cue integration. Two noisy cue streams (A and B) are presented simultaneously, each carrying partial information about one of 10 classes. A binary context signal indicates which cue is reliable on each trial (cue A or cue B). The fixed-pathway variant duplicates the context into both cue branches; the learned-routing variant requires the router to discover the cue separation from data. This task is used in Sec. 4.7 to test structured pathway-vector broadcast.

J Depth Scaling and Noise Robustness

Depth scaling. Varying dendritic depth from 1–4 layers (branch factors [9] to [3, 3, 3, 3]): shunting local degrades from 63.5% to 57.4%; additive local degrades more steeply from 54.9% to 29.7%. The shunting advantage grows with depth (+8.5 $\rightarrow$ +27.7 pp). Backprop ceilings remain stable at about 90–92%. See Fig. S6A.

Noise robustness. Gaussian noise $\mathcal{N}(0, \sigma^2)$ on broadcast error: shunting is robust to $\sigma \leq 0.1$ (about 62%) while additive drops from 46.5% to chance at $\sigma=1.0$, confirming that shunting credit signals carry useful learning information beyond the broadcast magnitude. See Fig. S6B.

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- The assumptions made should be given (e.g., Normally distributed errors).
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